
AIM:

To create a smart system that automates both **street lighting** and a **tollgate barrier** using sensors and electronic components.

Title:

Street Light with Tollgate

Apparatus / Components Required:

- Arduino Uno
 - Ultrasonic Sensor (HC-SR04)
 - Servo Motor
 - LEDs (for streetlight simulation)
 - Resistors
 - Jumper wires
 - Breadboard
 - Power supply or USB cable
-

Circuit Diagram:

(To be drawn manually or using simulation tools like Tinkercad/Fritzing)

- Ultrasonic Sensor:

- Trig → Pin 9
 - Echo → Pin 10
 - Servo Motor → Pin 3
 - LEDs can be connected to any digital pin (for street light simulation)
 - Common GND and power connections from Arduino
-

Working Principle:

- **Tollgate Section:**
 - The **ultrasonic sensor** continuously measures the distance of any approaching object.
 - If a vehicle is detected within 25 cm, the **servo motor** rotates to open the gate (90°).
 - After 3 seconds, the gate returns to its closed position (0°).
 - **Street Light (Simulated with LEDs):**
 - Can be added separately using LDR and LEDs, though not included in this specific code.
 - Automatically turns ON/OFF based on surrounding light intensity.
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Applications:

- Automatic Tollgate Systems
- Smart Highways
- Parking Lot Management

- Street Lighting Automation (if integrated with LDR)
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Performance Result:

- The system accurately detects vehicle presence using an ultrasonic sensor.
 - The servo motor responds effectively by opening and closing the gate automatically.
 - Reduces need for manual toll collection or gate operation.
 - System runs efficiently with minimal power consumption.
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Arduino Code:

```
// eazytronic.com - Toll Gate using Ultrasonic Sensor and Servo
#include <Servo.h> // Servo library
```

```
Servo s1;
```

```
// Pin Definitions
```

```
const int trigPin = 9; // Ultrasonic sensor trig pin
```

```
const int echoPin = 10; // Ultrasonic sensor echo pin
```

```
long duration;
```

```
int distanceCm, distanceInch;
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  s1.attach(3); // Servo connected to pin 3
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
}
```

```
void loop() {
```

```
  // Triggering the ultrasonic sensor
```

```
  digitalWrite(trigPin, LOW);
```

```
  delayMicroseconds(2);
```

```
  digitalWrite(trigPin, HIGH);
```

```
delayMicroseconds(10);  
digitalWrite(trigPin, LOW);
```

```
// Calculating distance  
duration = pulseIn(echoPin, HIGH);  
distanceCm = duration * 0.034 / 2;  
distanceInch = duration * 0.0133 / 2;
```

```
Serial.print("Distance: ");  
Serial.print(distanceCm);  
Serial.println(" cm");
```

```
// If vehicle is detected (distance < 25cm), open the gate  
if (distanceCm < 25) {  
    s1.write(90); // Open gate  
    delay(3000); // Wait for 3 seconds  
    s1.write(0); // Close gate  
    delay(1000); // Small delay to prevent re-triggering  
}
```

```
delay(200); // Small loop delay  
}
```
